STAFF REVIEW · H446 255 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. They share a memory pathway or resource.

2. An event such as a dependency, branch or memory stall disrupting instruction overlap.

3. The operation part of a machine or assembly instruction.

4. mid4 value10 → low5 high8, mid6 value14 → low7 high8, mid7 value16 → found. Indexes 4,6,7.

I Do — source worked steps

1. Predict first — Values are brought from memory into registers before the arithmetic instruction.

2. Explain and check — A regular load/store instruction set simplifies certain datapaths but may need more assembly instructions.

We Do — guided CHECK

1. Predict first
The complex ISA version uses fewer listed instructions but could require more work per instruction.
Explain and check
Evaluate clock cycles, decode overhead, code size, memory behaviour and hardware rather than choosing by instruction count alone.

You Do — source mark schemes

1. [4] RISC typically simpler/regular instructions [1] and load/store designs [1]; CISC permits richer complex instructions [1] and often variable instruction lengths or complex addressing [1].

2. [6] The compound instruction can express more work in one opcode [2]; RISC separates memory access and arithmetic [2]; number of instructions is not the same as measured execution time [2].

3. [6] Simpler regular instructions may support efficient decoding [1] and pipelining [1]; more instructions may increase code size [1] and fetch demand [1]; complex instructions can improve code density [1]; real performance depends on implementation/workload [1].

Exit Ticket — source answer

1. False. Instructions can differ in work and execution cost.